

HEILONGJIANG WANGYUNFENG WIND POWER PROJECT

Document Prepared By Longyuan (Beijing) Carbon Asset Management Technology Co.,Ltd.

Project Title	Heilongjiang Wangyunfeng Wind Power Project
Version	02
Date of Issue	03-04-2013
Project ID	UNFCCC Ref.5711
Monitoring Period	20/01/2010 to 30/01/2012
Prepared By	Longyuan (Beijing) Carbon Asset Management Technology Co.,Ltd.
Contact	Address: Floor7, No.6-9 International investment Building, Fuchengmen North Street, Xicheng District, Beijing, China. 100034 Tel: 86 10 6609 1020 Email: xuyang@clypg.com.cn

Table of Contents

1	PROJECT DETAILS	3
1.1	SUMMARY DESCRIPTION OF PROJECT	3
1.2	SECTORAL SCOPE AND PROJECT TYPE	4
1.3	PROJECT PROPONENT	4
1.4	OTHER ENTITIES INVOLVED IN THE PROJECT	4
1.5	PROJECT START DATE	4
1.6	PROJECT CREDITING PERIOD	4
1.7	PROJECT LOCATION.....	4
1.8	TITLE AND REFERENCE OF METHODOLOGY	4
2	IMPLEMENTATION STATUS	5
2.1	IMPLEMENTATION STATUS OF THE PROJECT ACTIVITY.....	5
2.2	DEVIATIONS FROM THE MONITORING PLAN	5
2.3	GROUPED PROJECT	5
3	DATA AND PARAMETERS.....	5
3.1	DATA AND PARAMETERS AVAILABLE AT VALIDATION	5
3.2	DATA AND PARAMETERS MONITORED.....	5
3.3	DESCRIPTION OF THE MONITORING PLAN.....	7
4	QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS	12
4.1	BASELINE EMISSIONS.....	12
4.2	PROJECT EMISSIONS	15
4.3	LEAKAGE	15
4.4	SUMMARY OF GHG EMISSION REDUCTIONS AND REMOVALS	15
5	ADDITIONAL INFORMATION	16

1 PROJECT DETAILS

1.1 Summary Description of Project

Heilongjiang Wangyunfeng Wind Power Project , (hereafter referred as the project) is located in Luobei County, Hegang City, Heilongjiang Province, China. Totally 58 sets of 850kW wind turbines were installed, providing a total capacity of 49.3MW. It is estimated the power supply was 120,069 MW annually and achieved amount of 123,430 tCO₂e emission reduction every year as per registered PDD. The purpose of the project is to generate power by wind and supply it to Northeast China Power Grid (NECPG).

The Project started construction on 04/06/2009, and the Project start date is 20/01/2010. The wind farm operated normally and smoothly. The project was registered as a CDM project on 31/01/2012.

The monitoring period of the project is from 20/01/2010 to 30/01/2012. The total emission reduction of the first monitoring period is: **195, 043 tCO₂e**.

Main technical information of wind turbines used in the project

Parameter	Value
Installed Capacity (kW)	850
Diameter of rotor (m)	G58-850/G52-850
Rated wind speed (m/s)	13
Cut-in wind speed (m/s)	3
Cut-out wind speed (m/s)	21
Survival wind speed (m/s)	37.5
Swept area (m ²)	2642/2214
Lifetime	20 year
Generator	Doubly fed asynchronous generator

1.2 Sectoral Scope and Project Type

The project is not a grouped project.

The Project comes under sectoral scope 1, Energy Industries renewable sources.

Project Category: Renewable electricity in grid connected applications

1.3 Project Proponent

.

Project Proponent	Hegang Longyuan Wind Power Co., Ltd.
Attention	Wang Zhewei
Title	Project manager
Direct Tel	0451 51919055
Email address	wangzhewei@clypg.com.cn

1.4 Other Entities Involved in the Project

No other entities.

1.5 Project Start Date

The Project Start Date is 20/01/2010, when the project started generating emission reductions.

1.6 Project Crediting Period

The Project crediting period is from 20/01/2010 to 30/01/2012, covering 2 years.

1.7 Project Location

The proposed project is located in Damahe Forest Farm, Luobei County, Hegang City, Heilongjiang Province, and P.R.China. Its geographical coordinates are north latitude 47° 54' 34' ' , and east longitude 130° 39' 56' ' , its altitude is about 240m~430m.

1.8 Title and Reference of Methodology

Title: Approved consolidated baseline and monitoring methodology ACM0002 "Consolidated baseline methodology for grid-connected electricity generation from renewable sources"- ACM 0002 (Version 12.1.0)

The methodology also refers to the latest approved versions of the following tools:

- ◆ Tool to calculate the emission factor for an electricity system (Version 02.2.1);
- ◆ Tool for the demonstration and assessment of additionality (Version 05.2.1).

2 IMPLEMENTATION STATUS

2.1 Implementation Status of the Project Activity

The Project was started construction on 04/06/2009, and the Project start date is 20/01/2010.
The electricity generated by the Project is delivered to NECPG.

2.2 Deviations from the Monitoring Plan

NA

2.3 Grouped Project

NA

3 DATA AND PARAMETERS

3.1 Data and Parameters Available at Validation

Data Unit / Parameter:	$EF_{grid,CM,y}$
Data unit:	t CO ₂ e/MWh
Description:	Baseline emission factor in the year y
Source of data:	Registered PDD
Value applied:	1.0280
Purpose of the data:	Baseline emission calculations.
Any comment:	The emission factor of the project ($EF_{grid,CM,y}$) is ex-ante determined and is fixed during the first crediting period.

3.2 Data and Parameters Monitored

Data Unit / Parameter:	$EG_{facility,y}$
Data unit:	MWh/yr
Description:	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y
Source of data:	Measured electricity meters installed in the substation
Description of measurement methods and procedures to be applied:	Project activity site
Frequency of monitoring/recording:	The reading of electricity meters are continuously measured and monthly recorded.
Value monitored:	Baseline emission calculations
Monitoring equipment:	See as follow table 1
QA/QC procedures to be applied:	Bidirectional electricity meters are used for monitoring the quantity of power exported and imported. Net electricity generation supplied can be obtained by the export subtract the import. The meters' accuracy is 0.5s, and are calibrated yearly according to national standards and rules.
Calculation method:	$EG_{facility,y} = E_{export,y} - E_{import,y}$
Any comment:	The designated personnel are responsible for recording the data. Sales receipts of the project is also obtained for cross check

Data / Parameter:	$EG_{export,y}$
Data unit:	MWh
Description:	Electricity exported to the grid by the project
Source of data:	Electricity meter reading on project activity site
Measurement procedures (if any):	The designated person of the project reads and aggregates the number of the meter. Data is archived for 2 years following the end of the crediting period by means of electronic and paper backup.
Monitoring frequency:	The reading of electricity meters are

	continuously measured and monthly recorded.
QA/QC procedures to be applied:	The meters' accuracy is 0.5S, and is calibrated yearly according to national standards and rules. The data is directly used to calculate emission reductions. The sales receipts are used to ensure consistency.
Any comment:	

Data / Parameter:	$EG_{import, y}$
Data unit:	MWh
Description:	Electricity utilized by the project
Source of data:	Electricity meter reading on project activity site
Measurement procedures (if any):	The designated person of the project reads and aggregates the number of the meter. Data is archived for 2 years following the end of the crediting period by means of electronic and paper backup.
Monitoring frequency:	The reading of electricity meters are continuously measured and monthly recorded.
QA/QC procedures:	The meters' accuracy is 0.5S, and are calibrated yearly according to national standards and. The data is directly used to calculate emission reductions. The sales receipts are used to ensure consistency.
Any comment:	

3.3 Description of the Monitoring Plan

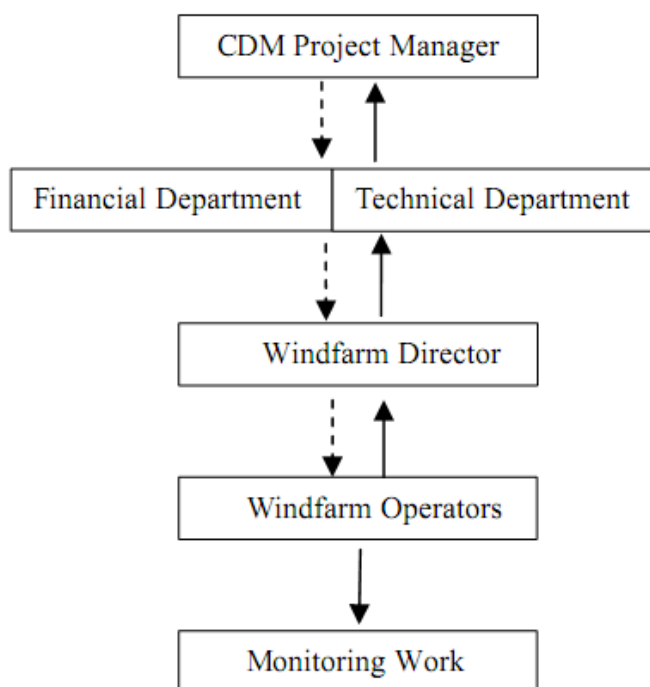
1. Introduction

The project adopts the approved consolidated baseline and monitoring methodology ACM0002 "Consolidated baseline methodology for grid-connected electricity generation from renewable

sources (Version 12.1.0)” to determine the emission reductions from the wind farm. This plan describes in more detail the process.

2. Responsibility

Hegang Longyuan Wind Power Co., Ltd. is overall responsible for monitoring and carries out the monitoring following this monitoring plan. The CDM manager of Hegang Longyuan Wind Power Co., Ltd. is responsible for the monitoring and reporting of the wind farm. The management structure is illustrated as follows:



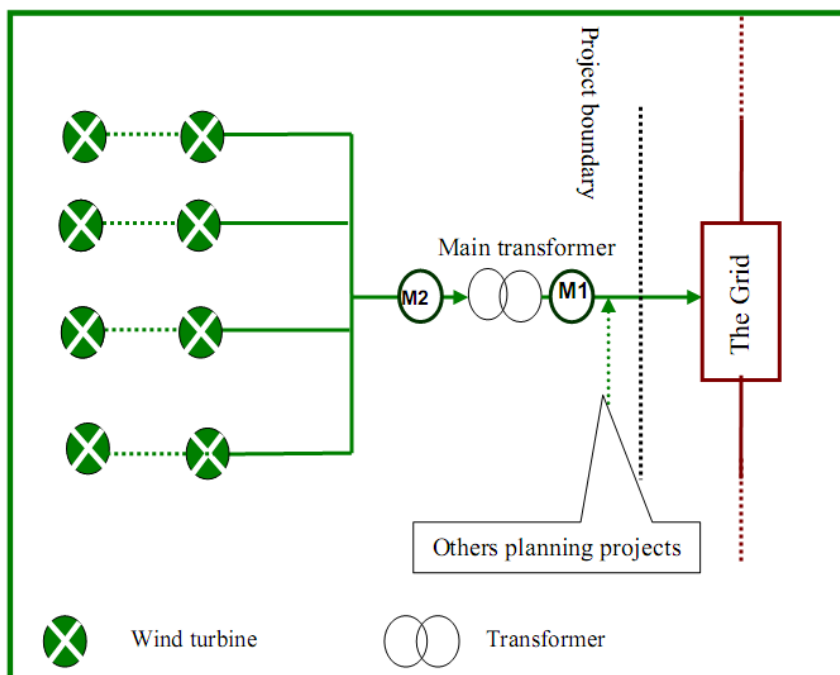
Windfarm Director collects the information and data required by the Monitoring Plan. The collected information is documented and sent to the CDM manager and responsible staffs of Hegang Longyuan Wind Power Co., Ltd. monthly. The CDM manager is in charge of the implementation of the Monitoring Plan and the confirmations on monitoring, calculation data and report to the General Manager of the company. Managers of the proposed project must maintain credible, transparent, and adequate data estimation, measurement, collection, and tracking systems to maintain the information required for an audit of an emission reduction project.

3. Training

Hegang Longyuan Wind Power Co., Ltd. assigns and trains the dedicated people carrying out the monitoring work. The monitoring personnel training by the contacted CDM consult company is completed before the registration, further training work is completed with the preliminary verification.

4. Meters system

For the proposed project, a meter (M1, accuracy degree is 0.5S, bidirectional) with a backup meter (M2, accuracy degree is 0.5S, bidirectional) is installed to monitor directly $EG_{import,y}$ and $EG_{export,y}$ by the proposed project (shown in the figure below).



5. Calibration

The metering equipments related with the proposed project are calibrated and tested yearly by a qualified third party appointed by the Northeast China Power Grid for accuracy according to the requirement from Technical administrative code of electric energy metering. The meters shall be jointly inspected and sealed on behalf of the parties concerned and shall not be interfered with by either party except in the presence of the other party or its accredited representatives.

Table 1 Monitoring Equipments of the Project

Meter	Serial number	Calibrate Date	Validity	Accuracy	Calibration Frequency	Calibration Entity
Main meter M1	4200525040909208867	20/05/2009 17/05/2010 15/05/2011 12/05/2012	yes	0.5s	Annually	Hegang Metrological Verification Institution
M2	4200222040909208865	20/05/2009 17/05/2010 15/05/2011 12/05/2012	yes	0.5s	Annually	Hegang Metrological Verification Institution

6. Monitoring data

According to the illustration figure above, the net electricity supplied to the grid by the proposed project is calculated by:

$$EG_{\text{facility},y} = EG_{\text{export},y} - EG_{\text{import},y}$$

7. Quality assurance and Quality control

The quality assurance and quality control procedures for the emergency about recording, maintaining and archiving data shall be improved as part of this CDM project activity. This is an on-going process that can be ensured through the CDM in terms of the need for verification of the emissions on an annual basis according to methodology ACM0002.

Hegang Longyuan Wind Power Co., Ltd specially issued the regulations of monitoring and management for CDM project (hereafter refer as CDM Manual) to monitor and verify the emission reductions from the proposed project. Moreover, the management procedures and regulations like operation, meter recordings, maintenance and emergency management are established in the CDM Manual, which guides the technicians to operate and guarantee the quality assurance and quality control procedures for recording, maintaining and archiving data in detail. This is an

on-going process that can be ensured through the CDM in terms of the need for verification of the emissions on an annual basis according to methodology ACM0002.

Several processes required to meet the requirements for emissions reduction monitoring include:

- The project owner reads the meters system monthly and supplies the recordings to the local Electric Power Company. The local Electric Power Company checks the recordings supplied by the project owner.
- If either party finds any reading of the Meters more than the allowable error or the monitoring equipment functions improperly, it should inform the other party immediately. The project owner and the local Power Grid Company should retain a qualified measure institute together to check the meters or equipment, solve the problems and get everything into normal condition; In the mean time, both sides shall jointly prepare an acceptable estimation method according to Regulations and Rules of Power Supply¹ issued by State Electricity Regulatory Commission of China to calculate the electricity exported to the Grid, and provide evidence to DOE for the verification to show the estimation is reasonable and conservative.

For the handling of disputes between the proposed project owner and the Grid, measures are adopted according to relevant articles of Interim Measures for Settlement of Electricity Charges between the Power Generating Enterprises and the Grid Enterprises² issued by State Electricity Regulatory Commission of China. Moreover, if meters are out of order, the electricity generation in the period is not considered.

8. Data Management System

Overall responsibility for monitoring greenhouse gas emissions reductions rests with the CDM monitoring staff of the proposed project. The CDM manual sets out the procedures for tracking information from the primary source to data calculations in paper format. Moreover, the credibility and reliability of those data and information must be confirmed. Physical documentation such as paper-based maps, diagrams and environmental assessment are collated in a central place, together with this monitoring plan. In order to facilitate auditor's reference, monitoring results are indexed. All paper-based information will be stored by Hegang Longyuan Wind Power Co., Ltd and kept at least one copy, and all data including calibration records is kept until 2 years after the end of the total crediting period of the CDM project.

¹ http://www.serc.gov.cn/flfg/bmgz/200802/t20080220_5931.htm

² <http://www.chinapower.com.cn/article/1130/art1130580.asp>

The following table below outlines the key documents relevant to monitoring and verification of the emission reductions from the proposed project.

Table List of the key documents relevant to monitoring and verification

I.D.No.	Document Title	Main Content	Source
F-1	PDD, including the electronic spreadsheets and supporting documentation (assumptions, estimations, measurement, etc)	Calculation procedure of emission reduction and monitoring items	Project owner or UNFCC website
F-2	Report on monitoring and checking of electricity supplied to the grid	Record based on monthly meter reading and electricity sale receipts	Project owner
F-3	Report on maintenance and calibration of metering equipment	Reasons for maintenance and calibration and the precision after maintenance and calibration	Project owner
F-4	Report on the qualifications of the operators	Technical post, working experience etc.	Project owner
F-5	the project management record (including data collection and management system)	Comprehensively and truly reflect the management and the operation of the proposed project	Project owner

9. Verification and monitoring results

The verification of the monitoring results of the proposed project is a mandatory process required for all CDM projects. The main objective of the verification is to independently verify that the project has achieved the emission reductions as reported and projected in the PDD. It is expected that the verification could be done annually.

3.4 Baseline Emissions

According to ACM0002 and the registered PDD of the Project, The baseline emission BE_y during the monitoring period results from:

$$BE_y = EG_{\text{facility},y} \times EF_{\text{grid},CM,y}$$

$$EG_{\text{facility},y} = EG_{\text{export},y} - EG_{\text{import},y}$$

Where :

BE_y is the baseline emissions of the Project;

$EF_{grid,CM,y}$ is the combined margin baseline emission factor of the NECPG;

$EG_{facility,y}$ is the net electricity supplied to the grid by the Project;

$EG_{export,y}$ is the quantity of electricity exported to the grid by the Project;

$EG_{import,y}$ is the quantity of electricity imported from the grid by the Project.

The monthly electricity data is listed in following table 2:

M1 is taken as the main meter of the project. Also, the grid company provides sales receipt to crosscheck the electricity that generated from the project. The electricity figures in table 2 are the most conservative ones by comparing the M1 monthly record and sales receipt from the grid company. For the comparing process, please check the project ER sheet.

Table2. Calculation of the net electricity supplied to the grid by the project

Period		Electricity exported to the grid by the project ($EG_{export,y}$) (MWh)	Electricity imported from the grid by the project ($EG_{import,y}$) (MWh)	The quantity of net electricity generation supplied by the Project (MWh)
From	To	A	B	C=A-B
20/01/2010	31/01/2010	1,309.440	4.620	1,304.820
01/02/2010	28/02/2010	3,784.440	50.160	3,734.280
01/03/2010	31/03/2010	8,649.960	35.640	8,614.320
01/04/2010	30/04/2010	8,914.620	11.220	8,903.400
01/05/2010	31/05/2010	7,288.380	14.520	7,273.860
01/06/2010	30/06/2010	4,492.620	31.680	4,460.940
01/07/2010	31/07/2010	3,746.820	48.180	3,698.640
01/08/2010	31/08/2010	5,790.840	15.840	5,775.000
01/09/2010	30/09/2010	8,755.560	21.780	8,733.780
01/10/2010	31/10/2010	9,327.120	15.180	9,311.940
01/11/2010	30/11/2010	10,722.360	31.680	10,690.680

01/12/2010	31/12/2010	9,109.980	62.700	9,047.280
Subtotal		81,892.140	343.200	81,548.940
01/01/2011	31/01/2011	13,377.540	19.800	13,357.740
01/02/2011	28/02/2011	8,490.900	26.400	8,464.500
01/03/2011	31/03/2011	9,119.220	11.220	9,108.000
01/04/2011	30/04/2011	7,305.540	25.740	7,279.800
01/05/2011	31/05/2011	6,717.480	20.460	6,697.020
01/06/2011	30/06/2011	4,996.200	24.420	4,971.780
01/07/2011	31/07/2011	2,558.160	35.640	2,522.520
01/08/2011	31/08/2011	4,386.360	43.560	4,342.800
01/09/2011	30/09/2011	8,696.160	11.220	8,684.940
01/10/2011	31/10/2011	10,416.780	8.580	10,408.200
01/11/2011	30/11/2011	10,862.940	13.860	10,849.080
01/12/2011	31/12/2011	10,649.100	31.680	10,617.420
Subtotal		97,576.380	272.580	97,303.800
01/01/2012	30/01/2012	10,922.340	43.560	43.560
Subtotal		10,922.340	43.560	10,878.780
01/01/2010	30/01/2012			189,731.520

The baseline emission during this monitoring period calculated as following:

$$\text{Total: } BE_y = EG_{\text{facility},y} \times EF_{\text{grid},CM,y} = 189,731.520 \text{ MWh} \times 1.0280 \text{ tCO}_2\text{e/MWh}$$

$$= 195,043 \text{ tCO}_2\text{e}$$

$$\text{Year 2010: } BE_y = EG_{\text{facility},y} \times EF_{\text{grid},CM,y} = 81,548.940 \text{ MWh} \times 1.0280 \text{ tCO}_2\text{e/MWh}$$

$$= 83,832 \text{ tCO}_2\text{e}$$

$$\text{Year 2011: } BE_y = EG_{\text{facility},y} \times EF_{\text{grid},CM,y} = 97,303.800 \text{ MWh} \times 1.0280 \text{ tCO}_2\text{e/MWh}$$

$$= 100,028 \text{ tCO}_2\text{e}$$

Year 2012: $BE_y = EG_{\text{facility},y} \times EF_{\text{grid},CM,y} = 10,878.780 \text{ MWh} \times 1.0280 \text{ tCO}_2\text{e/MWh}$
=11,183 tCO₂e

Vintage	$EG_{\text{facility},y}$ (MWh)	$EF_{\text{grid},CM,y}$ (tCO ₂ e/MWh)	BE_y (tCO ₂ e)
2010	81,548.940	1.0280	83,832
2011	97,303.800	1.0280	100,028
2012	10,878.780	1.0280	11,183
Total	189,731.520	1.0280	195,043

3.5 Project Emissions

Project emission (PE_y) is 0 tCO₂e as per the registered PDD.

3.6 Leakage

Leakage (L_y) is 0 tCO₂e as per the registered PDD.

3.7 Summary of GHG Emission Reductions and Removals

The total Emission Reduction (ER_y) during this monitoring period is calculated as follow:

$$ER_y = BE_y - PE_y - L_y = 195,043 \text{ tCO}_2\text{e} - 0 \text{ tCO}_2\text{e} - 0 \text{ tCO}_2\text{e} = 195,043 \text{ tCO}_2\text{e}$$

$$ER_{2010} = BE_{2010} - PE_{2010} - L_{2010} = 83,832 \text{ tCO}_2\text{e} - 0 \text{ tCO}_2\text{e} - 0 \text{ tCO}_2\text{e} = 83,832 \text{ tCO}_2\text{e}$$

$$ER_{2011} = BE_{2011} - PE_{2011} - L_{2011} = 100,028 \text{ tCO}_2\text{e} - 0 \text{ tCO}_2\text{e} - 0 \text{ tCO}_2\text{e} = 100,028 \text{ tCO}_2\text{e}$$

$$ER_{2012} = BE_{2012} - PE_{2012} - L_{2012} = 11,183 \text{ tCO}_2\text{e} - 0 \text{ tCO}_2\text{e} - 0 \text{ tCO}_2\text{e} = 11,183 \text{ tCO}_2\text{e}$$

Vintage	BE_y (tCO ₂ e)	PE_y (tCO ₂ e)	L_y (tCO ₂ e)	ER_y (tCO ₂ e)
2010	83,832	0	0	83,832
2011	100,028	0	0	100,028

2012	11,183	0	0	11,183
Total	195,043	0	0	195,043

4 ADDITIONAL INFORMATION

The actual emission reductions during this monitoring period are 195,043tCO₂e, which is less than the estimated value in the registered PDD⁵. There is no any significant increase compared with the estimated emission reduction in the registered PDD.

⁵ As per ex-ante calculation of the registered CDM-PDD, the average of annual emission reductions is 123,430tCO₂e, so the estimated ERs during this period are 250,580 tCO₂e (=123,430*741/365)